International Conference on Smart Communication and Sustainable Technologies : Submission (133) has been edited.

2 messages

Microsoft CMT <noreply@msr-cmt.org> Tue, 17 Mar, 2026 at 12:00 am
To: 22r21a04a1@mlrit.ac.in

Hello,

The following submission has been edited.

Track Name: ICSCST2026

Paper ID: 133

Paper Title: YOLOv7-Based Vehicle Detection for Adaptive Traffic Signal Control

Abstract:
One of the key problems in cities is traffic congestion, which leads to the expansion of travel time, fuel use, and pollution of the environment. This paper puts forward an AI-based smart traffic management system that will dynamically adjust the timing of the traffic lights according to the real-time traffic density that is computed by taking images from cameras at the points where the roads intersect. The suggested system is based on computer vision and deep learning methods, where a YOLOv7 object detection model is applied to the traffic images to accurately identify and classify vehicles. The number of vehicles that is identified in each lane is used in estimating the density of traffic and setting adaptive signal timing, whereby high traffic density has the highest green signal timing so as to minimize unwarranted wastage and maximize the efficiency of the flow. Moreover, there is an interface for traffic conditions and signal status visualization in real time, which allows effective monitoring of the traffic and manual intervention in cases of necessity. The system is tested on various traffic images under various conditions of congestion, and the results of experiments prove a significant decrease in the waiting time of vehicles and the general time of congestion in comparison with the standard fixed-time traffic signal control systems. The proposed solution is implemented using Python and provides a cost-efficient and scalable way of working with intelligent transportation systems and smart city traffic control.

Created on: Thu, 19 Feb 2026 04:32:43 GMT

Last Modified: Mon, 16 Mar 2026 18:30:13 GMT

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Secondary Subject Areas: Not Entered

Submission Files:
YOLOv7_Based_Vehicle_Detection_for_Adaptive (1).pdf (1 Mb, Mon, 16 Mar 2026 18:29:54 GMT)

Submission Questions Response:

1. Track
   Track 4 - Artificial Intelligence & Machine Learning for Smart Systems: AI/ML methods for perception, decision-making, and optimization.
2. Keywords
   Artificial Intelligence, Computer Vision, Object Detection, Traffic Signal Control, Intelligent Transportation System, Deep Learning.

Thanks,
CMT team.

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[Quoted text hidden]